

*Skip List – Insert Element – Time Complexity According to William Pugh
–Type B*

Average Case – Time Complexity by William Pugh

This explanation is flawless, rigorous, and beautifully structured. It captures the exact classical recurrence relation formulated by William Pugh (the inventor of Skip Lists) in his original 1990 paper.

$C(k)$

- *$C(k)$ is the average total number of steps needed to climb up `k` levels during the backward walk.*

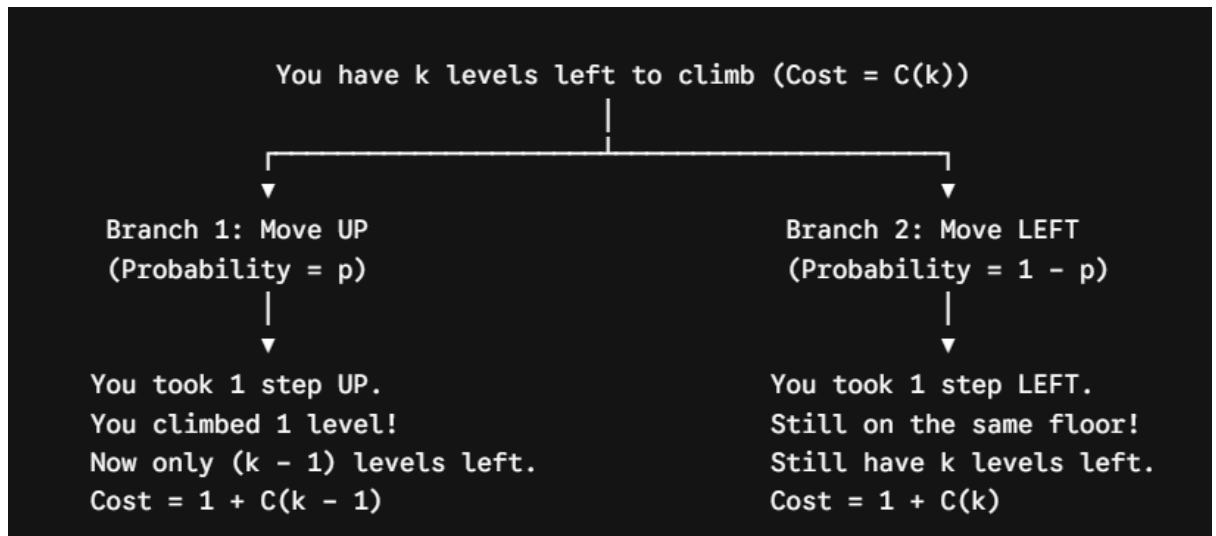
*To know what is `Backward Walk`, you have to go through **Type A** Average Case of Time Complexity of previous document.*

- *if we need to climb 5 levels, the cost is **$C(5)$** .*
- *if we need to climb 4 levels, the cost is **$C(4)$** .*
- *if we need to climb 0 levels, the cost is **$C(0) = 0$** (We are already at the top.)*

We are currently standing at a node and we have k levels left to climb.

We must take 1 step.

There are only two possible outcomes:



Explaining Each Branch

Branch 1: UP Branch $\rightarrow p(1 + C(k - 1))$

- **Probability (p)**: The current node exists on the level above.
- **The Step: 1 step UP**
- **Since we just climbed 1 level, we now only have $(k - 1)$ levels left to climb.**
- **Remaining Cost** : By definition , climbing $(k - 1)$ levels costs $C(k - 1)$.
- **Total Cost for this branch:**

$$\begin{array}{c}
 \text{1} \\
 \text{1 step UP}
 \end{array}
 +
 \begin{array}{c}
 C(k - 1) \\
 \text{Remaining levels}
 \end{array}
 = 1 + C(k - 1)$$

- *Weighted by probability p :*

$$p \cdot (1 + C(k - 1))$$

Branch 2: The Left Branch $\rightarrow (1 - p)(1 + C(k))$

- **Probability $(1 - p)$:** *The current node doesn't exist on the level above.*
- **The Step:** *We are blocked from going UP, so we take 1 step LEFT to the previous node.*
- **What remains?** *We did not climb any levels – we are still on the exact same floor! Therefore, we still have all the **k levels** left to climb.*
- **Remaining Cost:** *Climbing k levels from the new node still costs $C(k)$.*
- **Total Cost for this branch:**

$$\begin{array}{c}
 \text{1} \\
 \underbrace{\hspace{1cm}} \\
 \boxed{\text{1 step Left}}
 \end{array}
 +
 \begin{array}{c}
 C(k) \\
 \underbrace{\hspace{1cm}} \\
 \boxed{\text{still need } k \text{ levels}}
 \end{array}
 = 1 + C(k)$$

- *Weighted by probability p :*

$$(1 - p) \cdot (1 + C(k))$$

Now combining both the branches (Law of Total Expectation)

The total expected cost $C(k)$ is simply the weighted sum of both branches:

$$\text{Total Cost } C(k) = \text{Pr}(\text{LEFT}) \times (\text{Cost if LEFT}) + \text{Pr}(\text{UP}) \times (\text{Cost if UP})$$

- ***where `Pr` represents Probability.***

$$\begin{aligned}\therefore C(k) &= (1 - p)(1 + C(k)) + p(1 + C(k - 1)) \\ \Rightarrow C(k) &= (1 - p) + (1 - p) \cdot C(k) + p + p \cdot C(k - 1) \\ \Rightarrow C(k) &= (1 - p) + p + (1 - p) \cdot C(k) + p \cdot C(k - 1) \\ \Rightarrow C(k) &= 1 + (1 - p) \cdot C(k) + p \cdot C(k - 1)\end{aligned}$$

Subtracting $(1 - p) \cdot C(k)$ from both sides we get:

$$\begin{aligned}\Rightarrow C(k) - (1 - p) \cdot C(k) &= 1 + (1 - p) \cdot C(k) - (1 - p) \cdot C(k) + p \cdot C(k - 1) \\ \Rightarrow C(k) - (1 - p) \cdot C(k) &= 1 + p \cdot C(k - 1) \\ \Rightarrow C(k) - (C(k) - p \cdot C(k)) &= 1 + p \cdot C(k - 1) \\ \Rightarrow C(k) - C(k) + p \cdot C(k) &= 1 + p \cdot C(k - 1) \\ \Rightarrow p \cdot C(k) &= 1 + p \cdot C(k - 1)\end{aligned}$$

Dividing entire equation by p , we get:

$$\frac{p \cdot C(k)}{p} = \frac{1}{p} + \frac{p \cdot C(k - 1)}{p}$$

$$\Rightarrow C(k) = \frac{1}{p} + C(k - 1)$$

The formula : $C(k) = \frac{1}{p} + C(k-1)$ says:

*The cost to climb `k` levels is the cost of climbing **1** level $\left(\frac{1}{p}\right)$ plus the cost of climbing the remaining **k - 1** levels.*

Unrolling it all the way down to $C(0) = 0$, the topmost level:

By substitution method:

$$C(k) = \frac{1}{p} + C(k-1) \rightarrow k \text{ levels}$$

$$\text{When } C(k-1) = \frac{1}{p} + C(k-2)$$

Substituting $C(k-1) = \frac{1}{p} + C(k-2)$ in $C(k)$ we get:

$$C(k) = \frac{1}{p} + \underbrace{\left(\frac{1}{p} + C(k-2)\right)}_{\boxed{C(k-1)}}$$

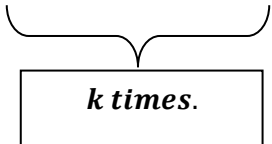
$$\text{When } C(k-2) = \frac{1}{p} + C(k-3)$$

Substituting $C(k-2) = \frac{1}{p} + C(k-3)$ in $C(k)$ we get:

$$C(k) = \frac{1}{p} + \left(\frac{1}{p} + \left(\frac{1}{p} + C(k-3) \right) \right)$$

Thus, repeating all way down to $C(0)$:

$$C(k) = \frac{1}{p} + \frac{1}{p} + \frac{1}{p} + \cdots + \frac{1}{p} + C(0)$$



*As from **Type A** Average Case of Time Complexity, previous document , we have seen there are ' $k+1$ ' level .*

We know, $C(0) = 0$, applying we get:

$$\Rightarrow C(k) = \frac{1 + 1 + 1 + \cdots k \text{ times}}{p} + 0$$

$$\Rightarrow C(k) = \frac{1 \times k}{p}$$

$$\Rightarrow C(k) = \frac{k}{p}$$

*We know from , **Type A** Average Case of Time Complexity Analysis, that*

$$MAX_LEVEL = \log_{\frac{1}{p}} n$$

Where **MAX_LEVEL** is the maximum height of the list.

When climbing all $k = \log_{\frac{1}{p}} n$ levels of the Skip List:

$$C(k) = \frac{k}{p}$$
$$\Rightarrow C\left(\log_{\frac{1}{p}} n\right) = \frac{\log_{\frac{1}{p}} n}{p}$$

For $p = 0.5$:

$$\Rightarrow C\left(\log_{\frac{1}{0.5}} n\right) = \frac{\log_{\frac{1}{0.5}} n}{0.5}$$

$$\Rightarrow C\left(\log_{\frac{1}{\frac{1}{2}}} n\right) = \frac{\log_{\frac{1}{\frac{1}{2}}} n}{\frac{1}{2}}$$
$$\Rightarrow C(\log_2 n) = 2 \times \log_2 n$$

And we get $O(\log_2 n)$, when $p = 0.5$ falls in $O(\log n)$ class.

And both notations represent the exact same complexity class.

$$O(\log_2 n) \approx O(\log n).$$
